

# JEE MAIN REVISION MASTER: CHEMICAL EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 11 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                                | Frequency | Category              |
|---------------|-------------------------------------------------------------|-----------|-----------------------|
| High          | Equilibrium Constants ( $K_p$ , $K_c$ ) & Reaction Quotient | 8 PYQs    | Equilibrium Constants |
| Moderate      | Le Chatelier's Principle & Stress Factors                   | 3 PYQs    | Le Chatelier          |

## 2. Master Formula Reference Card

| Concept / Topic                                             | Mathematical Formulation & Governing Equations                                                                             |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| Equilibrium Constants ( $K_p$ , $K_c$ ) & Reaction Quotient | $K_p = K_c(RT)^{\Delta n_g}$ , $\Delta G^\circ = -RT \ln K = -2.303RT \log_{10} K$ ,<br>$Q < K \implies \text{Forward}$    |
| Le Chatelier's Principle & Stress Factors                   | $\ln \left( \frac{K_2}{K_1} \right) = \frac{\Delta H^\circ}{R} \left( \frac{1}{T_1} - \frac{1}{T_2} \right)$ (van 't Hoff) |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Equilibrium Constants ( $K_p$ , $K_c$ ) & Reaction Quotient (High Yield — 8 PYQs)

**Core Principle:** Relationship between  $K_p$  and  $K_c$ , expression of equilibrium constants, and spontaneous direction based on  $Q$  vs  $K$ .

■ **Caution / JEE Pitfall:** In calculating  $\Delta n_g$ , count ONLY gaseous species:  $\Delta n_g = \sum n_{\text{(gas)}}(\text{products}) - \sum n_{\text{(gas)}}(\text{reactants})$ . Pure solids and liquids are excluded ( $a = 1$ ).

■ **Speed Trick:** If a reaction is multiplied by factor  $n$ , its new equilibrium constant becomes  $K' = K^n$ . If reversed,  $K' = 1/K$ .

**Representative Source PYQs:**

- [2025 Shift] Q#1 (MCQ): "The equilibrium constant at 298 K for the reaction  $A+B \rightleftharpoons C+D$  is 100. If the initial concentrations of all the four species are 1.0 mol/L, the reaction will proceed in the forward direction." [Ans: N/A]
- [2025 Shift] Q#3 (MCQ): "Which of the following statements is/are true about equilibrium? (a) Equilibrium is possible only in a closed system of at a given temperature." [Ans: N/A]
- [2023 Shift] Q#2 (MCQ): "At 500K, for a reversible reaction  $A_2(g) + B_2(g) \rightleftharpoons 2AB(g)$  in a closed container,  $K_c = 2 \times 10^{-5}$ . In the..." [Ans: N/A]

### 3.2 Le Chatelier's Principle & Stress Factors (Moderate Yield — 3 PYQs)

**Core Principle:** Dynamic response of equilibrium systems to variations in concentration, temperature, pressure, and inert gas addition.

■ **Caution / JEE Pitfall:** Addition of an inert gas at CONSTANT VOLUME has NO effect on equilibrium position because partial pressures remain unchanged!

■ **Speed Trick:** Inert gas addition at CONSTANT PRESSURE shifts equilibrium toward the side with MORE gaseous moles ( $\Delta n_g > 0$  shifts forward).

**Representative Source PYQs:**

- **[2026 Shift] Q#1 (MCQ):** "For the following gaseous reversible reaction:  $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_3\text{B}_{(g)}$  ( $\Delta_r H = -q$ )..." **[Ans: N/A]**
- **[2026 Shift] Q#2 (MCQ):** "For the reversible reaction,  $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons 2\text{NH}_3(\text{g})$  When the partial pressure is measured i..." **[Ans: N/A]**
- **[2025 Shift] Q#4 (MCQ):** "According to Le Chatelier's principle, in the reaction  $\text{CO}(\text{g}) + 3\text{H}_2(\text{g}) \rightleftharpoons \text{CH}_4(\text{g}) + \text{H}_2\text{O}(\text{g})$ , the formation of methane i..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

For the following gaseous reversible reaction:  $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_3\text{B}_{(g)}$  ( $\Delta_r H = -q$  kJ). The amount of product  $\text{A}_3\text{B}_{(g)}$  is affected by \_\_\_\_

- (A) Temperature alone
- (B) Pressure alone
- (C) Both temperature and pressure
- (D) Temperature, pressure and catalyst

**Correct Answer: None**

**Solution Analysis:** For the reaction  $3\text{A}_{(g)} + \text{B}_{(g)} \rightleftharpoons \text{A}_3\text{B}_{(g)}$  the forward reaction is exothermic because  $\Delta_r H = -q$  kJ. So, heat is released in the forward direction. If temperature is increased, the equilibrium shifts in the backward direction to absorb heat. So the amount of  $\text{A}_3\text{B}$  decreases. If temperature is decreased, the equilibrium shifts in the forward direction. So the amount of  $\text{A}_3\text{B}$  increases. Now look at pressure. On the reactant side, the number of gaseous moles is  $3+1=4$ . On the product side, the number

### Benchmark Problem 2: KCET 2025 (Q#1)

The equilibrium constant at 298 K for the reaction  $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$  is 100. If the initial concentrations of all the four species were 1 M each, then equilibrium concentration of D (in  $\text{mol L}^{-1}$ ) will be

- (A) 0.182
- (B) 1.818
- (C) 1.182
- (D) 0.818

**Correct Answer: None**

**Solution Analysis:**  $\text{A} + \text{B} \rightleftharpoons \text{C} + \text{D}$   
 $1-x \quad 1-x \quad 1+x \quad 1+x$   
 $100 = \frac{(1+x)(1+x)}{(1-x)(1-x)} \Rightarrow (1+x / 1-x) = 10 \Rightarrow 1+x = 10-10x \Rightarrow 11x = 9$   
 $x = (9 / 11) = 0.818$  (D)  $1+x = 1+0.818 = 1.818$